MASTER UPDATE PROMPT

Hospital Management System (Nepal) - Complete Feature Update

Project Information

I already have a Hospital Management System built in Django.

Do NOT rebuild the project from scratch.

Instead:

- Analyze the existing project completely.
- Analyze every module.
- Analyze the GitHub repository structure.
- Reuse the existing template.
- Keep the existing design language.
- Improve the project professionally.
- Maintain consistency across all pages.
- Improve UI/UX.
- Improve security.
- Improve database relationships if needed.
- Do not remove existing features unless they conflict with these requirements.

The project is for a hospital in Nepal.

Everything should look professional like a real hospital ERP.

PRIMARY GOAL

I want this system to become a complete Hospital ERP.

The UI must be clean.

The workflow must be simple.

Everything must connect together.

Every department should communicate with each other through the patient record.

The system must be modular.

Everything must be role-based.

Everything should be searchable.

Everything should be filterable.

Everything should be secure.

IMPORTANT

Before writing any code:

1. Analyze the whole project.
2. Analyze database models.
3. Analyze permissions.
4. Analyze routing.
5. Analyze templates.
6. Analyze dashboard.
7. Analyze GitHub repository again.
8. Reuse existing components.
9. Keep code clean.
10. Do not duplicate logic.

USE EXISTING TEMPLATE

Continue using the same UI template already used in the project.

Do NOT replace it.

Improve it only.

Maintain same color scheme.

Maintain responsive design.

DJANGO ADMIN

Install and configure Django Jazzmin (or Jasmine if already intended).

Make Django Admin look modern.

Inside Super Admin Dashboard add:

- Django Admin

When clicked,

Open Django Admin panel.

Configure Jazzmin professionally.

Customize:

- Hospital logo

- Sidebar
- Colors
- Icons
- Branding

ROLE BASED ACCESS CONTROL (RBAC)

Implement strict Role-Based Access Control.

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1. Super Admin

Super Admin has full access.

Can:

- View everything
- Edit everything
- Delete everything
- Print everything
- Configure system
- Manage users
- Manage permissions
- View accounts
- View reports
- Access Django Admin
- Restore deleted records
- Audit logs
- Override permissions

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2. Registration Counter

Can:

Register patients

Generate:

- Patient ID
- OPD Ticket

Search patient by:

- Name
- Phone
- Hospital ID
- QR Code

View:

- Daily registrations
- Monthly registrations
- Yearly registrations

Can:

Edit registrations

Delete registrations

ONLY within 24 hours.

After 24 hours:

Editing automatically becomes disabled.

Only Super Admin can modify.

Registration Dashboard

Display cards:

Today's Registration

This Month

This Year

Collection Today

Collection Month

Collection Year

If number is large:

Display:

1.2 lakh

instead of

120000

When clicked:

Open popup dialog.

Show detailed report.

Close button available.

Do NOT open another page.

Registration Form

Required:

First Name

Last Name

Gender

Phone Number

Address

Age

Age Type

Age Type dropdown:

Years

Months

Weeks

Days

Example:

24 Years

3 Months

2 Weeks

10 Days

OPD Ticket

New Patient

100 NPR

Old Patient

50 NPR

Automatically detect new or old patient.

Payment Method

Default:

Cash

Dropdown:

Cash

eSewa

Future ready for:

- FonePay
- Khalti
- Card

Doctor Module

Doctor can:

Search patient.

Open patient profile.

View:

History

Lab Reports

Radiology

Medical Records

Pharmacy

Admission

Operation Theater

Blood Bank

Insurance

Doctor Notes

Visit History

Doctor can:

Add Notes

Prescriptions

Diagnosis

Treatment Plan

Recommend Admission

Recommend Surgery

Request:

Lab

Radiology

Blood

Medicine

Doctor Dashboard

Show:

Patients Today

This Month

This Year

Filter:

Daily

Weekly

Monthly

Yearly

Custom Date

Doctor Quota System

Very important.

Doctor can create OPD quota.

Example:

Dr. Sharma

Monday

50 Patients

Tuesday

40 Patients

Wednesday

70 Patients

Registration counter must follow quota.

If quota is full:

Cannot register more patients.

Quota appears automatically in Registration.

Cash Counter

Search patient.

Add services quickly.

Smart search.

Example:

Type:

X

Shows:

X-Ray

Type:

E

Shows:

ECG

Type:

U

Shows:

Ultrasound

Type:

B

Blood Test

Everything searchable.

Autocomplete.

Keyboard friendly.

Cashier Dashboard

Cards:

Today's Collection

Monthly Collection

Yearly Collection

Popup details

Date filter

Department filter

Patient filter

Receipt history

Laboratory

Upload report.

Upload PDF.

Scan PDF.

Automatically attach to patient Medical Record.

Search patient.

View previous reports.

Print reports.

Download PDF.

Cannot edit after verification.

Radiology

Same workflow as Laboratory.

Upload:

PDF

Image

Automatically save into Medical Record.

Medical Record Department

Medical Record is centralized.

Everyone can access according to permission.

Only Admin can edit.

Everyone else:

Read only.

Access:

Doctor

Registration

Nursing

OT

Blood Bank

Admission

Insurance

Pharmacy

Can View

Cannot Edit

Cannot Delete

Can Print Reports

Cannot Print Patient Card

Pharmacy

View prescriptions.

Medicine sales.

Filter:

Today

Month

Year

Patient

Medicine

Revenue

Nursing

Search patient.

View:

Medical Record

Lab

Radiology

Admission

Doctor Notes

Vitals

Cannot delete.

Cannot edit reports.

Admission

View beds.

Bed availability.

Occupied.

Discharged.

Admitted today.

Patient search.

Medical record access.

Operation Theater

Access:

Complete patient history.

Medical records.

Lab

Radiology

Admission

Blood

Doctor Notes

Operation Notes

Can upload OT reports.

Blood Bank

Search patient.

Blood request.

Blood issued.

Blood history.

Medical Record.

Insurance

Nepal Government Insurance support.

Private Insurance support.

QR Scanner.

Insurance Number Entry.

If QR fails:

Manual entry.

Insurance Approval.

Insurance Discount.

Claim Tracking.

Patient can view insurance history.

Insurance Claim Receipt

Generate A4 receipt.

Contains:

Patient Details

Insurance Number

Patient QR

Insurance QR

Services

Discount

Remaining Amount

Hospital Amount

Claim Amount

Print

PDF

JPG

Save into Medical Record.

Accounts Module

View revenue.

Department-wise revenue.

Daily

Monthly

Yearly

Custom Date

Registration Revenue

Laboratory Revenue

Radiology Revenue

Pharmacy Revenue

Admission Revenue

Insurance Revenue

Grand Total

Super Admin can drill down.

Click department.

View all transactions.

Patient Portal

Patient Login must NOT be inside Admin Dashboard.

Create separate public website.

Homepage:

Hospital Information

Patient Login

Sign Up

Forget Password

Patient Registration

Phone Number

OTP Verification

Email Optional

Password

Confirm Password

Profile Photo

Address

Gender

Age

Emergency Contact

Hospital ID

Automatically generated.

Online registration and physical registration must use same sequence.

Example:

Online:

HT-001

Counter:

HT-002

Online:

HT-003

No duplicate IDs.

Patient Dashboard

View:

Medical Records

Lab Reports

Radiology

Insurance

Prescriptions

Visit History

Appointments

Downloads

Can download:

PDF

JPG

Cannot print Patient Card.

Only Registration and Super Admin can print Patient Card.

Appointment System

Patient books appointment.

Doctor quota checked automatically.

Available slots only.

Integrated with Registration.

Search System

Every module should search by:

Hospital ID

Phone

Name

QR Code

Patient Number

Insurance Number

QR Code

Generate patient QR.

Scan anywhere.

Open patient profile.

Reports

Every report downloadable:

PDF

JPG

Print

Analytics

Every department should have dashboard cards.

Examples:

Today's Revenue

Monthly Revenue

Yearly Revenue

Patients Today

Pending Reports

Completed Reports

Cards should open popup.

Popup contains full details.

UI Requirements

Professional ERP style.

Responsive.

Clean.

Fast.

Bootstrap.

Modern Icons.

Smooth animation.

Popup Modal.

Large numbers:

1.2 lakh

3.5 crore

instead of raw values.

Security

Use Django authentication.

Role-based permissions.

Audit logs.

CSRF protection.

Input validation.

SQL Injection protection.

XSS protection.

Secure password hashing.

Phone verification.

Nepal Localization

Use Nepal-specific data.

Examples:

Kathmandu

Pokhara

Lalitpur

Bhaktapur

Province

Municipality

Ward Number

NPR currency.

Date formatting suitable for Nepal.

Printing Rules

Only:

Registration

Super Admin

Can print Patient Card.

Everyone else:

Can print reports only.

Final Task

Before finishing:

- Review every module.
- Ensure all modules communicate correctly.
- Ensure Medical Record is the central source of truth.
- Ensure permissions are correct.
- Ensure patient privacy is maintained.
- Ensure UI is consistent.
- Ensure code is optimized.
- Remove duplicate code.
- Fix bugs.
- Improve performance.
- Maintain clean Django architecture.
- Add comments where necessary.
- Follow Django best practices.
- Ensure production-ready quality.

Expected Deliverables

The update should include:

Module	Status
Registration	Enhanced
Patient Portal	Enhanced
Doctor	Enhanced
Laboratory	Enhanced
Radiology	Enhanced
Pharmacy	Enhanced

Module	Status
Nursing	Enhanced
Admission	Enhanced
Operation Theater	Enhanced
Blood Bank	Enhanced
Insurance	Enhanced
Cash Counter	Enhanced
Medical Records	Centralized
Accounts	Advanced
Reports	Advanced
Dashboard	Professional
Role Permissions	Fully Implemented
QR System	Implemented
Doctor Quota	Implemented
Online Registration	Connected
Popup Analytics	Implemented
Jazzmin Admin	Implemented
Security	Hardened
Nepal Localization	Implemented

Development Guidelines

Category	Requirement
Framework	Django (reuse existing project)
Database	Reuse existing database, migrate safely
Frontend	Continue existing template and Bootstrap UI
Admin	Django Jazzmin customization
Authentication	Django Auth with Role-Based Access Control

Category	Requirement
Media	PDF, JPG uploads and downloads
QR	Generate and scan patient QR codes
Search	Autocomplete, smart search across modules
Payments	Cash (default), eSewa (ready for future gateways)
Reporting	PDF export, print-friendly layouts
Audit	Activity logs for sensitive actions
Localization	Nepal-specific addresses, NPR currency, hospital workflow
Performance	Optimize queries, avoid duplicate logic, use reusable components
Security	CSRF, XSS, SQL injection protection, secure password storage, permission checks

Goal: Transform the existing Django Hospital Management System into a production-ready, secure, modular, Nepal-focused Hospital ERP while preserving the current architecture and template, improving functionality rather than rebuilding it.

Okay, and yes, there are different modules of such a managing system I can see this. And laboratory, they can upload a PDF or scan a PDF, auto-sign it to medical record. That is good. And in radiology also same workflow. Okay. They can search the patient. And in the cashier, cashier counter they can do all the bill payment, blood tests, sugar test, and this X-ray, that, this all, whatever the office pay, all they are done by this. And medical record, okay, that is also a very nice thing. It will be only accessed by the few. Super admin can delete, edit anything, but medical record can be viewed by the different people. Doctor. Doctor can view the medical report. Registration can view the medical report. Admin can do the edit anything, you know that. And nursing can view the report, all. They can't edit, but they can view the report. Operation theater can, blood bank can, and medical record itself is a medical record. And the medical record account is also with the admin. It's a different account, but most probably it will be used by the doctor, admission nurse, operation theater, blood bank. That's what I mean, okay. And okay, and there was a problem I saw there, which you need to make to fix. Let's come to the role-based access metrics. Super admin, all, everything they can do. They can have everything. They can do everything, everything, everything, everything. Registration counter, yeah, register patient ID card, OPD ticket, view patient, view how much they have, and how much the bill they have generated today. And it can be seen by the time, setting of time, and it can be seen of three years ago also. And altogether bill also, in rupees, Nepali rupee, ru-... and NRP-... It should be systematic and you know what I mean. You can make it, I know. And search patient history, yeah. And doctor is accessed with the, there should be a medical records option also, where the doctor can scan and just see the medical report. Cash counter, yeah. Patient billing, all the X-ray, ECG, everything. And the system should look very perfect. If I have an option like, if I go for the patient billing and search a patient and now there the option billing, okay, click, then there should be option like multiple, it might be of 100 or anything. And when I type X, then X-ray, E then ECG, U then ultrasound, some kind of options like that, it should be there. For all this, this should be in registration counter for the address search. And in make sure that same thing in pharmacy, insurance, admission, nursing, operation theater, blood bank, account, revenue, everywhere. Doctor can see how many patients they have this day, this month, this year, by time, setting of time. And one thing you have to insert that, the quota system. For example, the doctor has access of providing the quota for the limited people. For example, the doctor from urology can see 100 patients from 5 to 10, example. Then he can make that quota. So there should be making a quota option. Similarly, similarly, okay. And that quota will be directly appeared in the registration counter. So the registration counter person cannot just simply register whatever they want. If they have quota, then they can. If they don't have, they can't. So it should be like that. And for registration, name, last name, age, gender, phone number, address is compulsory. And the age can be of 24 years old, 24 months, 24 days, 24 weeks, many things it can be. So there should be option. And while paying, new ticket 100 rupees, old ticket 50 rupees. It should work like that. You know what I mean. And make sure the option is there of payment when there is cash and e-sewa. Cash and e-sewa. Wherever the option, primary is cash, default cash, and e-sewa is optional. And yeah, for registration, there is a option, for example, it can be of staff will get the 90% off. And the insurance, and insurance, yeah, okay, I'll tell about insurance also. Laboratory, okay, that's all good. And in laboratory also they can see the record. And each, these things have their access of patient. And patient has access of seeing the medical record of their own. They can't see of others. They can download the reports, but they can't download their card, patient card they can't download. So they can download their reports in PDF, JPG, something, download PDF. And for everywhere there should be a back logo where they will be directly, they can go back one one step and go one one step forward. One one step forward they can go by clicking, but one one step back is, we need some kind of thing, okay. And yeah, that's all. Do you know one thing? There should be like one working mechanism. For example, doctor also have another feature to admit the patient, some kind of thing. Okay. Search patient, then the patient will appear there and he can do like history, notes, lab report, radiology, surgery, admit, and pharmacy, and operation theater, blood bank. That's all his access and medical report he can see, all this. But one thing, make sure the patient is accessed by all, but they can be seen only their thing. Blood bank will see the medical record of blood bank only. Other will not be opened. Operation theater can see all, all. Give them access, operation theater doctor. Nurse can also see all. And admin also can see all. Registration counter can also see all. Doctor also can see all. And super admin have the access to see all and edit, delete as well as. But others don't have the option of editing or deleting. But simply they have the option of to print, not the patient ID card, but the report only. Okay? And that's the things and account, yeah, make sure it is like similarly, systematic way, sorting by the date, month, year. We can see of one year, two year, three year, 2020, 2021, and it will be appeared like that. And similarly, there will be like that day patient, how many were there, and which counter generated what, what, what, what. And registration this much, not the least exactly, but this much of data, this much of data, this much of data, this much of data is collected. For example, 10,000 is collected from the cash counter. Similarly, some kind of thing. Make it so systematic and everything good, okay. But the super admin have access to see all if they want to see the accounts and the report. And they can see in accounts or some kind of accountant things, and they can see when they click for a account and registration, how many patient, all the list of patient. It's like a big thing, okay. Clicking the date and they can see. And how much amount he generated that day. Similarly, some kind of, some for similar pharmacy incidents admission. Super admin is super powerful here. And yes, there should be a booking appointment type of options in the dashboard, first of all in the home page, what I mean. And there should be a simple kind of login, patient login. Make sure it is good. And when they log in, they will directly make their profile, good their profile. And there they can see their ID card, medical history, all, everywhere, like doctor, patient, everything, everything they can see. And they can just view their report. And there should be another appointment. Make sure the appointment is not in the dashboard, make sure it's appointment inside the patient registration. Okay. And if the patient is new, create one, then they automatically get their new patient number. They need to enter the phone number, and they can enter that phone number is verified in their phone, and they can enter the email or the phone number, and they can create the password two times, and create one and then there will be a sign up. Similarly in that same box and add down, it will be sign up, and they can sign up, and for the sign up they just can do paste their hospital ID. And hospital ID and they can put the password and phone number, and they will be logged in. And inside logging their profile, they will be like, they should put all the information, and there it will be a registration option as well as, and the registration option has just connected with the similarly the registration counter. But this time the information will be same to same, same billing, same OPD ticket, but it's like if the patient pay with the cash, then cash payment, but from online they can't pay cash, so cash is pending should be there. And e-sewa, and similarly for registration new hundred and for old fifty, some kind of thing. And make sure this is all accessed by the super admin. And a patient can amount putting information, all these things, accessed by the super admin. And similarly we have, similarly we have like that. And forget password they can do. Make sure the patient can forget password and forget password recovery option is from only phone number. Phone number is primary, so give the phone number correct. It's kind of what I mean. I'm not saying to you, but phone number is like that. And going to there and they can, they have a profile, they can add a photo, similarly as a registration, they need to fill all the information, and they will get their card and for that card they need to visit the registration. And registration counter will only print that, and super admin will print that. Other, nobody have right to print patient card. Okay? And for me, what I want is if the person who was creating account from home is HT-001 and who got token 1, and just after that the patient who was in the registration counter registered and that person will be H-1-002 and token 2. Similarly, online and registration counter are related with each other. Okay? Make sure that. The token system we have given like you previously generated in OPD ticket, from registration by the patient from home, and this from here it will be like some kind of thing. And for registration, for patient make there the patient login option or the patient something type of things. Do not need to make it inside the dashboard, okay? Make it in an official website. And so it's a Nepali website, so information should be like related in Nepali. Not Nepali language, but Nepali information. For example, Kathmandu, then K-A-T-H-M-A-N-D-U. Similarly, what I mean, you understand this clearly. And yes, make sure if I have missed many of the activities, make sure it is very secure and very private. And make sure the password is password which you have created, one, two, three, star, that is perfect. Username should be that only. And whenever the super admin login, there should be another option like Django admin or Django web or Django option, Django admin page, something kind of in a dashboard. And whenever he clicks, he will go directly to the Django, and there the Django website should be using the Jam file or make it like a good Django. What do we call, what do we use the library called Django something? Make it. As I was saying the Django dashboard, make it the Jasmine. You have created the CSS and that okay, but there is an option. Like whenever we scroll down, there is an option going like registration counter, laboratory, pharmacy, nursing, operation, blood bank, account, and Django admin. And clicking it will direct to Django admin. And there the, you make it perfect and good looking with the Jasmine, what we call it, Jasmine or whatever I think. You make it, I mean, make it perfect, okay? And make sure the system is very well worked and it is perfect, perfect everywhere. Make sure you use the same template I gave you previously from the GitHub repository. Check it once again. Make it once again. Make a perfect, similar function should be there like I gave you in the repository. Analyze it again. See it again, okay? And make sure who have access to what, make sure that. And that's all. Yeah, that's all. Make it a perfect, make it a perfect, perfect, perfect. Whenever we go for a payment in anywhere, there should be option, primary cash and drop down then e-sewa. And for anything which have more information, there will be seen like a box. For example, imagine there is a report or there is a receipt or there is a total collection. Example for the registration counter, registration counter they can see their collection today, okay? And if it is 100000000, then simply write 1.1 lakh, example. And clicking that box will make it big. Not like a full window, but it will open like a big, something like a box. And there will be this, this, this collected, and the cross option in the box, crossing, then again moving. You know what I mean. Make sure the register can see their information like that. And they can see the total generated this month, today, year by clicking the option. And if the number is exceeding the box, make sure make it short. 100000000 means 1.1 lakh. And clicking it will make it a big one, similarly. Similarly like a box, simply, you know what I mean, make it systematic and there is a detailed information and we can cross that and it will again back that. It is not a next window, it is simply a popping up and making it big with many information. That's all the feature. Cash counter also same, they can see their total, what they have done. And make sure there is a system, it is compulsory that you have to make a system. Registration counter can edit, delete, modify the tickets, anything within a 24 hour. If not done within a 24 hour, it will be directly moved to the super admin and registration counter don't have access to do that. Okay. And regarding other things, you know that. And what I wanted to say is, yeah, in insurance. So insurance is also good. There should be options, how many insurance you done, and pharmacy also view the pharmacy sells, similarly by age, by patient. I'm not saying age, okay. By patient, by year, month, something like that. By patient also there. And patient have QR code, so make the system work with the QR code. There should be like, if the QR code is scanned, then that's all. Or putting the phone number or the name or the hospital ID, that would be perfect. Blood bank also okay. Account, yeah, I have already told. And nursing also they can see the patients, search the patient, patient history, search patient. Admission also similarly, how many bed admitted, whatever things had, and the medical record they can view. And for insurance, it is a Nepal insurance, and in Nepal there is a card which have QR code. So make sure there is government insurance system as well as, just make it like a prototype or something like where you can scan the QR code and it will directly take back to the official side of the government. But make sure it's not make like very advanced, make sure it works simply. If not, it directly, that's okay. But make sure it work like that. Or example, scan then his information we can add afterwards that. Whenever the scan, then or enter number is also there. Every time the scan doesn't work, okay? And different insurance system from different insurance company also there. You know what I mean, how to make it, you know all, okay? And yeah, and the insurance record if in Nepal there is an insurance record like government is paying the insurance of 4000 and they can, they only have access to use only 1 lakh for year, per year, for their family. So the patient insurance record, that is different, the patient can't see anything, okay? Okay, patient can see their insurance record as well as, okay. The patient can see. And insurance if one time is scanned and they will get like this percent off for cash counter and for radiology, pharmacy, admission, they only approve the things, okay? Scan and approve, and they'll get the option, but they'll scan and approve, and it will be like how many off, for example, 1 lakh, they will get like 80% off, I guess, 80-90. I don't know, make it simply like that, okay? How many percent off, make it. And they will be like approve, and there will be like print receipt, and then the things will be like patient ID, patient information, patient phone number, everything, and two QR code, patient QR code, and the insurance QR code, and the insurance number, and at the down like simply there will be X-ray insurance claimed, total this much, after this this much. Make sure these things all in one A4 size paper, and make sure it is similarly as a billing, and that can be downloaded in a PDF or a JPG. Okay, and that is as seen by this, that will be also added in a medical record. Insurance also can, and there is update, upload or something kind of things that can be added to the medical record as well as, if they want, it's not a compulsory. And patient can see their insurance detail as well as, but they can't, they can download but they can't download the patient ID. Patient ID is simply a unique ID which will be in like a ID card or the visiting card, simply small one, not the big A4 size paper, and it will be like printing with the JPG, PDF, or it can print from with the sticker, you know. Simple, same size of sticker will be printed, and there are some printer we can print. So, different thing. Make it so systematic and make it good.

make a prompt to update some things like this... in my project.... ok.. Explainsin details for easy understand still i miss details.... Andi don't net khalti and other, make it just text, oney eswa and cash, and cash primarily... you know what i mean.... And make postgare sql as well and squl3(whatever the latest andffast) for now squal withbbei and update read me, and give the details now

That's a good direction. Your previous prompt is already strong, but it still misses many technical implementation details that an AI developer needs to build a professional Hospital ERP. I would also avoid asking it to support multiple payment gateways if you only want **Cash** and **eSewa** for now.

I would rewrite the project requirements into a proper Software Requirements Specification (SRS)-style prompt so the AI understands exactly what to build, how the modules connect, what permissions each role has, how the database should work, and what should be updated in the README.

Below is a much more detailed version you can give to Cursor AI, Claude, ChatGPT, GitHub Copilot, or any coding AI.

HOSPITAL MANAGEMENT SYSTEM

COMPLETE PROJECT UPDATE PROMPT

Django + PostgreSQL + SQLite + Bootstrap

Nepal Hospital ERP

Version 2.0

IMPORTANT

This is **NOT a new project.**

This project already exists.

Your task is to **analyze, improve, optimize, refactor, and extend the existing project**.

Do **NOT** recreate the project.

Do **NOT** remove existing functionality unless absolutely necessary.

Instead:

- Analyze the existing codebase.
- Analyze the GitHub repository completely.
- Analyze every Django app.
- Analyze models.
- Analyze templates.
- Analyze routing.
- Analyze permissions.

- Analyze JavaScript.
- Analyze CSS.
- Analyze reusable components.
- Preserve the current design.
- Improve code quality.
- Improve database structure where necessary using migrations.
- Reuse existing components whenever possible.

PROJECT GOAL

Transform the existing Hospital Management System into a complete **Hospital ERP** suitable for hospitals in Nepal.

The system should be:

- Professional
- Fast
- Secure
- Modular
- Easy to maintain
- Responsive
- Production-ready

TECHNOLOGY STACK

Backend

- Python 3.13+ (latest stable supported by Django)
- Django (latest stable LTS)
- Django REST Framework (where appropriate)
- PostgreSQL (Primary Production Database)
- SQLite (Development Database)

The project must support **both** databases through Django settings.

- settings/development.py → SQLite
- settings/production.py → PostgreSQL

The database backend should be switchable using environment variables.

DATABASE REQUIREMENTS

Use PostgreSQL as the primary production database.

SQLite should remain available only for development and testing.

Optimize:

- Indexes
- Foreign keys
- Constraints
- Query performance
- Relationships

Avoid duplicate data.

Use proper normalization.

README UPDATE

Completely rewrite the README.md.

It should include:

Project Overview

Features

Screenshots (placeholders)

Installation

SQLite Setup

PostgreSQL Setup

Environment Variables

Database Migration

Create Superuser

Run Development Server

Production Deployment

Django Jazzmin Setup

Folder Structure

User Roles

Module Explanation

API (if available)

Backup and Restore

Security Features

Troubleshooting

Future Improvements

The README should be beginner-friendly and professionally formatted.

PAYMENT SYSTEM

For now, support **ONLY**:

1. Cash (Default)
2. eSewa

Do **NOT** integrate:

- Khalti
- FonePay
- Stripe
- PayPal
- Cards

Every payment form must default to:

Cash

Dropdown options:

- Cash
- eSewa

If payment is made online through the patient portal:

Cash should display as:

Pending Cash

until collected at the hospital (if applicable), or eSewa should mark the payment as completed based on your workflow.

ROLE-BASED ACCESS CONTROL (RBAC)

Implement strict permission-based access.

Super Admin

Full access:

- View
- Create
- Edit
- Delete
- Restore
- Export
- Print
- Configure
- Audit
- Django Admin
- System Settings
- User Management
- Database Backup

Admin

Can manage all hospital operations except critical system configuration and Django administration.

Registration Counter

Can:

- Register patients
- Generate Hospital ID
- Generate OPD Ticket
- Print Patient Card
- Search patients
- View medical records (read-only)
- Edit registrations within 24 hours
- Delete registrations within 24 hours
- View registration revenue

Cannot:

- Edit after 24 hours

- Delete after 24 hours
- Access Django Admin

Doctor

Can:

- View complete patient history
- View lab reports
- View radiology reports
- View prescriptions
- Add notes
- Prescribe medicine
- Request lab tests
- Request radiology
- Admit patient
- Schedule surgery
- Create OPD quota

Cannot:

- Delete records
- Edit completed reports
- Print patient ID card

Laboratory

Can:

- Search patient
- Upload PDF reports
- Upload scanned reports
- Attach reports automatically to Medical Records
- Print reports
- Download reports
- View previous laboratory history

Cannot edit verified reports.

Radiology

Same permissions and workflow as Laboratory.

Nursing

Can:

- Search patients
- View medical records
- View doctor's notes
- Record nursing observations
- View lab and radiology reports

Cannot delete records.

Admission

Can:

- Admit patient
- Assign bed
- Transfer bed
- Discharge patient
- View admission history
- View medical records

Operation Theater (OT)

Can:

- Access all patient medical information required for surgery
- View history
- View lab reports
- View radiology
- Add operation notes
- Upload OT reports

Blood Bank

Can:

- Search patient
- Manage blood requests
- Record blood issue
- View blood history

Only Blood Bank records are editable by Blood Bank staff.

Insurance

Supports Nepal Government Health Insurance and private insurance.

Functions:

- Scan QR (prototype/manual workflow)
- Enter Insurance Number
- Approve claim
- Apply discount
- Generate insurance receipt
- Save claim in Medical Record

Patients can view their own insurance history.

Accounts

Can:

- View revenue
- Department-wise reports
- Daily, monthly, yearly summaries
- Export reports
- Filter by department and date

Patient Portal

Patients can:

- Register
- Login

- Reset password (phone number only)
- Update profile
- Book appointments
- View medical records
- View insurance history
- View prescriptions
- Download reports (PDF/JPG)
- View OPD history

Patients **cannot**:

- Print patient ID card
- View other patients' data
- Edit medical records

PATIENT REGISTRATION

Required fields:

- First Name
- Last Name
- Gender
- Date of Birth or Age
- Age Value
- Age Unit (Years / Months / Weeks / Days)
- Phone Number (Primary)
- Address
- Municipality/City
- District
- Province
- Emergency Contact (optional)

Automatically generate:

- Hospital ID (e.g., HT-000001)
- QR Code
- OPD Token

Online registrations and counter registrations must share the same sequential numbering.

Example:

HT-000001 (Online)

HT-000002 (Counter)

HT-000003 (Online)

No duplicate IDs.

OPD TICKETS

New Patient:

100 NPR

Old Patient:

50 NPR

Automatically determine whether the patient is new or returning.

DOCTOR QUOTA

Doctors can define:

- Date
- Time Slot
- Maximum Patients

Registration staff can only book available slots.

No overbooking.

SMART SEARCH

All major modules should support searching by:

- Hospital ID
- Patient Name
- Phone Number
- QR Code
- Insurance Number

Service selection (Cash Counter, Billing, etc.) should support autocomplete. For example:

- Typing **X** suggests X-Ray.

- Typing **E** suggests ECG.
- Typing **U** suggests Ultrasound.
- Typing **B** suggests Blood Test.

MEDICAL RECORDS

The Medical Record module is the central source of truth.

All reports (Lab, Radiology, OT, Insurance, etc.) should be attached to the patient's timeline automatically.

Permissions:

Role	View	Add	Edit	Delete
Super Admin	Yes	Yes	Yes	Yes
Admin	Yes	Yes	Yes	No
Doctor	Yes	Notes/Orders	No	No
Registration	Yes	No	No	No
Laboratory	Own Reports	Yes	Before Verification	No
Radiology	Own Reports	Yes	Before Verification	No
Nursing	Yes	Nursing Notes	No	No
OT	Surgical Records	Yes	No	No
Blood Bank	Blood Records	Yes	No	No
Patient	Own Records	No	No	No

DASHBOARDS

Every department should have a dashboard with summary cards such as:

- Today's Activity
- Monthly Activity
- Yearly Activity
- Revenue (where applicable)
- Pending Tasks

Large numbers should be abbreviated for display (e.g., **1.2 lakh**). Clicking a card should open a modal dialog with detailed information instead of navigating to a new page.

SECURITY

Implement:

- Django authentication
- Strong password hashing
- CSRF protection
- XSS protection
- SQL injection prevention (through Django ORM)
- Audit logging for sensitive actions
- Role-based permissions
- Input validation
- Secure file uploads
- Environment variable management for secrets

DJANGO ADMIN

Install and configure **Django Jazzmin**.

Customize:

- Hospital logo
- Branding
- Sidebar icons
- Dashboard
- Colors

Add a "Django Admin" shortcut in the Super Admin dashboard that opens the Jazzmin interface.

LOCALIZATION

Target users are hospitals in Nepal.

Use:

- NPR currency
- Nepal addresses
- Provinces
- Districts

- Municipalities
- Appropriate date and number formatting

FINAL QUALITY CHECKLIST

Before considering the project complete:

- Analyze all existing modules.
- Remove duplicate code.
- Optimize database queries.
- Verify role permissions.
- Ensure Medical Records act as the central record.
- Test all workflows.
- Update documentation.
- Add comments where helpful.
- Keep the existing UI consistent.
- Ensure the application is production-ready.

This prompt should give an AI coding assistant enough detail to understand not just **what** to implement, but also **how the system should behave**, how the modules interact, which database strategy to use, what payment methods to support, how permissions should work, and how to update the project's documentation for future developers.